

Causal contour deformation

FastSecDec · fixed and smooth dynamic strength

Working report · 10 October 2026 · measurements not yet complete

1 Deformation styles and formulae

1.1 Common geometry and fixed strength

After sector mapping, separate the real endpoint monomials from the smooth density. Let $F(x)$ denote the designated residual causal polynomial and let $U(x)$ denote each retained positive residual factor. The definition of F includes its physical scale: rescaling F also rescales this deformation. For a sampled sector with real masses, real kinematics and $x \in [0, 1]^n$ ($n \geq 1$), define

$$w_i = x_i(1 - x_i), \quad v_i = w_i \partial_i F, \quad z_{i(x)} = x_i - i\lambda(x)v_{i(x)}.$$

Exact zero-dimensional contributions bypass sampling. The coordinate faces remain fixed in their normal direction. With $A = \sum_i w_i (\partial_i F)^2 \geq 0$, the ray obeys

$$\text{Im } F(x - itv) = -tA + O(t^3).$$

Thus a sufficiently small positive strength moves the causal denominator to the lower half-plane. The scalar λ is a strength, not the norm of the displacement; no undocumented normalization by F or its gradient is used. Since cube coordinates are dimensionless, $[\lambda] = [F]^{-1}$; the same units apply to L . The displacement limit R is in parameter-space units.

Fixed mode takes a finite runtime value $\lambda(x) = \lambda_0 > 0$. Its Jacobian is

$$J_{ij} = \delta_{ij} - i\lambda_0 \partial_j v_i, \quad \partial_j v_i = \delta_{ij}(1 - 2x_i) \partial_i F + w_i \partial_j \partial_i F.$$

The complex, oriented determinant $\det J$ multiplies the density. A zero determinant alone is not a pole crossing. Neither its absolute value nor a frozen-gradient approximation gives the same contour integral.

For endpoint exponents $a_i + b_i \varepsilon$, the real endpoint monomials are retained and the smooth density receives

$$\det J \prod_i \rho_i^{a_i + b_i \varepsilon}, \quad \rho_i = \frac{z_i}{x_i} = 1 - i\lambda(x)(1 - x_i) \partial_i F.$$

The last expression defines ρ_i even at $x_i = 0$. Deform the numerator and residual factors as well. Apply this pullback **before** Taylor/IBP subtraction and Laurent construction. Boundary terms restrict the same full-sector map; they do not construct a new, lower-dimensional deformation.

1.2 Branches and checking policy

Continue each factor separately. Endpoint ratios have real part one; positive residuals must keep the appropriate branch, and the causal logarithm has the lower-lip value $\log(F - i0) = \log(|F|) - i\pi$ when $F < 0$. Combining these continued logarithms into the principal logarithm of their product can change the sheet. A stationary point with $F \neq 0$ is allowed. A surviving singularity at $F = 0$ and $A = 0$ requires an unresolved-deformation diagnostic, not a fabricated zero; a natively cancelled denominator or finite smooth density remains finite.

Optional validation has three policies: **always** checks pilot and production, **pilot** checks separate pilot work and removes optional production checks, and **off** removes both. Solver termination and nonfinite handling remain active. Fixed-mode failures identify the factor and point; they never silently shrink the requested strength. Finite pilots are safeguards, not global certificates. The gradient-map convention and subtraction ordering follow the established sector-decomposition approach; see pySecDec.

1.3 Dynamic strength: a smooth sufficient causal radius

Set $\lambda(x) = Sr(x)$ with $0 < S < 1$. The radius bounds the **whole homotopy** $x - itv(x)$ for $0 < t \leq Sr(x)$. It is not the distance to the nearest complex root: a positive real ray root need not exist, and a root-modulus prescription can vanish at the very physical pole one must avoid.

In the auxiliary ray expansion, hold $v(x)$ fixed. For odd $k \geq 3$, define

$$T_k = D^k F(x)[v^{k-2}, \cdot, \cdot], \quad B_k = \frac{\sum_{i,j} w_i w_j (T_k)_{ij}^2}{(k!)^2}.$$

For every positive residual U , write the real even-ray expansion

$$\operatorname{Re} U(x - itv) = U(x) + \sum_{k \geq 2, \text{ even}} d_{k(x)} t^k, \quad C_k = \frac{d_k^2}{U(x)^2}.$$

Native polynomial positivity admission supplies a strictly positive residual U on the closed cube; an arbitrary positive floor is not substituted. Let m_F and m_U count the structurally retained odd and even orders. With positive cap L and displacement scale R , solve

$$H(x, r) = \frac{r^2}{L^2} + r^2 \frac{\|v\|^2}{R^2} + m_F \sum_{k \geq 3, \text{ odd}} B_k r^{2k-2} + \sum_U m_U \sum_{k \geq 2, \text{ even}} C_k r^{2k} = 1.$$

All coefficients are nonnegative. The cap term makes the positive root unique and simple, with $r \leq L$ and $rH_r \geq 2$ at the root. Both constraints are smooth; there is no hard minimum whose derivative would jump.

Why the bound is causal. Put $W = \operatorname{diag}(w_i)$ and $q = \sqrt{W} \operatorname{grad} F$, so $A = q^T q$ and $v = \sqrt{W} q$. The higher odd ray term satisfies

$$\left| D^k \frac{F[v^k]}{k!} \right| \leq A \sqrt{B_k}, \quad \sum_k \sqrt{B_k} t^{k-1} \leq \sqrt{m_F \sum_k B_k t^{2k-2}}.$$

Consequently higher imaginary terms cannot cancel the leading $-tA$ along the admitted homotopy when $A > 0$. The even-factor bound similarly keeps $\operatorname{Re} U(z) > 0$. The proof does not divide by A , so small gradients do not create an artificial singular bound. At $A = 0$, the stationary-point rules still apply.

1.4 Full local-strength derivatives

Dynamic strength changes the Jacobian:

$$J_{ij} = \delta_{ij} - i(\lambda \partial_j v_i + v_i \partial_j \lambda).$$

Implicit differentiation of $H(x, r(x)) = 1$ gives

$$\partial_j r = -\frac{\partial_j H}{\partial_r H}, \quad \partial_j \lambda = S \partial_j r.$$

Here $\partial_j H$ holds r fixed but differentiates all physical coefficient dependence, including v and the smooth constraints. Symbolic endpoint reduction differentiates the complete deformed density, including this identity and the determinant. Solver iteration choices are never differentiated. Every face retains the full-sector dimension, structural counts and cap parameters. Omitting the rank-one term $-iv(\operatorname{grad} \lambda)^T$ changes the integral.

The polynomial construction is an independently checked baseline. Its exact-arithmetic proof is distinct from numerical validation of a rounded implementation. Disabling optional checks removes their runtime cost, not solver failures or nonfinite-value diagnostics.

1.5 Sign-aware refinement of the dynamic bound

The baseline bounds beneficial as well as harmful higher terms. A smoother, tighter alternative bounds only terms that can oppose causality. Define the real symmetric matrix

$$R_k = \frac{(-1)^{\frac{k+1}{2}} L^{k-1}}{k!} \sqrt{W} T_k \sqrt{W}.$$

It need not be built with explicit square roots of W . Weighted trace and squared-norm identities provide the native polynomial quantities

$$a_k = \frac{\text{tr}(R_k)}{n}, \quad b_k^2 = \frac{n-1}{n} \left(\text{tr}(R_k^2) - \frac{\text{tr}(R_k)^2}{n} \right).$$

For a symmetric matrix its largest eigenvalue is at most $a_k + \sqrt{b_k^2}$. Smooth both the spectral bound and its positive part with a fixed dimensionless $\delta = 10^{-3}$:

$$t_k = a_k + \sqrt{b_k^2 + \delta^2}, \quad p_{\delta(t)} = \frac{t + \sqrt{t^2 + \delta^2}}{2}, \quad \mu_k = p_{\delta(t_k)}.$$

For negative t , evaluate the exactly equivalent expression

$$p_{\delta(t)} = \frac{\delta^2}{2(\sqrt{t^2 + \delta^2} - t)}$$

to avoid cancellation. For each even U coefficient, use

$$\nu_{U,k} = p_{\delta(-d_k \frac{L^k}{U})}.$$

With $u = \frac{r}{L}$, form

$$E_{F(u)} = \sum_{k \geq 3, \text{ odd}} \mu_k u^{k-1}, \quad E_{U(u)} = \sum_{k \geq 2, \text{ even}} \nu_{U,k} u^k.$$

Then solve the unique positive root

$$u^2 + \frac{L^2 u^2 \|v\|^2}{R^2} + E_{F(u)}^2 + \sum_U E_{U(u)}^2 = 1, \quad \lambda = S L u.$$

The nonnegative envelope coefficients retain monotonicity and smoothness. Their derivatives, including smoothing, enter the same implicit-jet machinery. A larger permitted displacement is a diagnostic; it does not establish lower variance or lower time to precision. The baseline and sign-aware construction must be compared with their full evaluation costs.

1.6 Sampling, diagnostics and interpretation

Korobov remains an outer real map: $y = T(q)$ followed by $z = z(y)$. Its total Jacobian is $\det J_{z(y)} \prod_i T_{i'}(q_i)$, with each factor applied once. No inverse Korobov derivative is introduced at an endpoint. Contour evaluation is deterministic and consumes no production random streams.

QMC uncertainties come from complete independently randomized lattices; democratic QMC retains covariance between sectors sharing a shift. Discrete Havana MC retains its frozen production proposal and uses complete batches. Both accumulate the full Laurent vector and its covariance. A partial allocation, a missing sector or an unvisited zero is not exact evidence.

The smooth sufficient-radius bounds above are FastSecDec's parameter-space construction. They are not attributed to pySecDec or to the LTD algorithm. Related work includes Winterhalder et al., *Targeting Multi-Loop Integrals with Neural Networks* on optimized contour choices, and Capatti et al., *Numerical Loop-Tree Duality: contour deformation and subtraction* on causal contour construction and physical multiloop tests. Their timings are not directly comparable FastSecDec speed measurements.

2 Physical $gg \rightarrow hh$: the 1000 GeV double box

The input is the existing individual **D05** s-channel double-box contribution, with incoming ++ helicities, $\sqrt{s} = 1000$ GeV, $\cos(\theta) = \frac{4}{5}$, $m_H = 125$ GeV and $m_t = y_m = 172.5$ GeV. HEPKit supplies the native graph, contracted numerator, wavefunctions and kinematics. The colour contraction is unnormalized δ_{ab} ; the measure is $\prod_l d^D \frac{k_l}{i\pi^{\frac{D}{2}}}$, $D = 4 - 2\varepsilon$, with multiplier one. This is not a complete diagram sum or a spin/colour averaged matrix element.

Scope correction: symbolic endpoint comparison pending. The generation measurements below use numerical-dual endpoint reduction. They are historical diagnostics. The requested comparison will use symbolic endpoint reduction with either symbolic or contour-only dual Jacobians; old high-order endpoint jet costs do not establish limitations of Jacobian-only dualization.

2.1 Generation and resident programs

Implementation	Generation / peak RSS	Sampling tradeoff
Fixed, symbolic determinant	84.2 s / 14.17 GB	Pending measurement
Dynamic, symbolic determinant	1104.4 s / 59.64 GB	Pending measurement
Native dual determinant option	Source-sector test only	See section 3

Both runs use sector-at-a-time serial generation, numerical-dual IBP, Horner iterations zero, common-pair rounds 1000 and SymJIT O2. Fixed uses eight workers; dynamic uses four. The 30 stochastic sectors plus 30 exact records occupy 79.04 MB and 175.08 MB respectively. Times include publication. The dynamic coordinator remains approximately 11 MB; completed mapped expressions stay on disk. The aggregate parent-and-child RSS cap is **100 GB** (decimal).

2.2 Five minutes on 50 physical cores

Method	Contour	Finite coefficient	Joint variance	Relative SE
QMC	Fixed	Pending measurement	Pending measurement	Pending measurement
QMC	Dynamic	Pending measurement	Pending measurement	Pending measurement
Discrete MC	Fixed	Pending measurement	Pending measurement	Pending measurement
Discrete MC	Dynamic	Pending measurement	Pending measurement	Pending measurement

For the finite complex estimate $\hat{I}_0$, let Σ_0 be its real-imaginary covariance matrix. The reported joint estimator variance is $V = \text{tr } \Sigma_0$ and relative joint standard error is $\frac{\sqrt{V}}{|\hat{I}_0|}$. This is not pointwise integrand variance. Retain both diagonal entries and the off-diagonal covariance in the accompanying reproducibility record.

The 300-second budget uses native integration elapsed time, including worker context setup and Havana adaptation. Artifact restoration, runtime binding and the separate causal pilot are reported outside that clock. Record actual elapsed time, cancellation drain, accepted replicas/batches, incomplete work, sample count, per-sector mean/maximum sample costs and peak RSS. Use independent production streams, a frozen 50-core affinity and separate tuning seeds. Fixed and dynamic settings are chosen in bounded preliminary runs and remain unchanged during production. No independent numerical reference exists for this point: finite error bars alone do not establish physical accuracy.

3 Implementation and performance

3.1 Preserve algebraic structure and reuse native facilities

FastSecDec owns the sector/integration interfaces in Rust. Symbolica supplies factored atoms, differentiation, function maps, evaluator optimization, duals, and serialization; Numerica supplies arithmetic, matrices, RNG primitives and Havana. HEPKit and Linnet retain graph, model and kinematic ownership. HEPKit's Python bridge wraps native objects rather than duplicating algorithms.

The smooth density is deformed before subtraction. Native function definitions can share repeated Jacobian and envelope bodies while preserving their derivatives and explicit radius dependencies. The saved program restores the optimized evaluator; binding a strength or cap must not rerun Horner/CPE. SymJIT O2 remains the native backend default; portable exact evaluator IR is retained separately for eager and high-precision execution.

An actual D05 source sector contains 236 smooth terms. The initial representation copied its large Jacobian into each term. Native structured construction plus a shared determinant body reduced that sector's mapped record from 683.8 MB to 365.6 MB and sampled peak RSS from 2.178 GB to 1.288 GB. Mapping wall time was 10.93 s versus 7.95 s on a shared host. In the native debug graph, compilation fell from 169.49 s to 58.16 s. Saved full-vector parity passes at two admitted points; both representations reject the same excessive strength. Saved program size increases by 6.6%. These are one-sector measurements, not full-generation timings or a convergence result.

3.2 Symbolic versus dual determinant

The symbolic route constructs a native determinant and optimizes it together with the density. It offers common-expression elimination across outputs but may make generation expensive when a large body is repeatedly substituted. **Historical implementation.** The first optional dual route composed native first-order image duals with a determinant program and then applied outer numerical-dual subtraction to the whole density. It preserved derivatives but coupled the two choices, contrary to the clarified requirement. Its source-zero dynamic program occupies 2.16 MB versus 5.74 MB, while compilation takes 190.55 s versus 91.30 s; fixed programs occupy 1.14 MB versus 2.64 MB. These measurements include high-order endpoint jets and cannot establish the cost or limits of contour-only Jacobian dualization.

Required independent construction. Keep endpoint reduction and subtraction symbolic in both variants. Native differentiation must retain every bulk, boundary, determinant and local-strength derivative. The proposed lowering uses first-order native image duals only for surviving Jacobian-entry expressions; the higher derivatives already generated by symbolic IBP remain symbolic. Restrict derivatives to a face only after differentiating the original map. No change of mathematical ordering is assumed. The capability probe, complete scientific checks and matched generation/RSS/program-size/sampling-cost comparison are pending. No alternative CAS or AD implementation is introduced.

3.3 Residency, callback ownership and statistical work

Universal indexed artifacts let serial generation publish one completed sector and release its worker process. Selective loading retains optimized native programs. Ordinary discrete MC uses caller-owned parallel execution; the 50-core campaign must account for all resident evaluator contexts. Aggregate parent-and-child RSS measures actual residency, including JIT scratch.

Independent probes exposed a native JIT-clone callback issue: cloned evaluator tables were not used by machine code, which retained the original callback environment and serialized its mutable workspace. The narrow owner correction, Symbolica PR 62, is retained in public consumer revision 74225696. The same consumer adds Symbolica PR 63: common-subexpression lookup remaps operands before hashing them, avoiding one optimizer pass per level of a duplicate dependency chain. Native branch ancestry and callback argument order remain unchanged. The FastSecDec workspace passes 917 tests (33 existing diagnostics ignored), portable tests pass 82 controls, and binding checks and strict workspace Clippy pass.

The physical QMC run also exposed avoidable coordinator work: matching common shifts used linear searches within already-sorted native replica records. Binary search preserves the selected rows and arithmetic order. Cancellation now drains dispatched results without repeating expensive observational reductions and checkpoint writes for every return; the final native checkpoint and report still include accepted returns. Deterministic cancellation and reordered, gapped-shift covariance/restoration controls cover these changes.

Stream identities are coordinator-owned. Pilots, production replicas and retries keep separate identities; rejected, duplicate or stale work cannot enter the estimate. Optional causal checks are truly absent in unchecked production, while solver/nonfinite failures remain visible. Accepted common QMC shifts and complete MC batches update the full covariance, preserving sector cancellations even at a time-budget stop.

Reproduction and source: FastSecDec, contour_deformation. Native input: examples/contour/gghh_double_box_1000/. Mathematical audit: docs/reviews/contour-foundation.md. Final campaign provenance and validated implementation revision will accompany the completed measurements.